1. $\lim\limits_{t\to 0}\left(1^{\frac{1}{\sin^2 t}}+2^{\frac{1}{\sin^2 t}}+.....+n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ is equal to : **[JEE(Main)Jan.2023]**

$\lim\limits_{t\to 0}\left(1^{\frac{1}{\sin^2 t}}+2^{\frac{1}{\sin^2 t}}+.....+n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{(n+1)}{2}$ (B) n^2 (C) n^2+n (D) n

Ans. D

Sol. $\lim\limits_{t\to 0}\left(\frac{1}{n^{\sin^2 t}}\right)^{\sin^2 t}\left[1+\left(\frac{1}{n}\right)^{\text{cosec}^2 t}+\left(\frac{2}{n}\right)^{\text{cosec}^2 t}+.....\left(\frac{n-1}{n}\right)^{\text{cosec}^2 t}\right]^{\sin^2 t}$

as $0<\frac{1}{n}<1$, So $\left(\frac{1}{n}\right)^{\infty}=0$

So, $n\,[1+0]^0 = n$

2. The set of all values of a for which $\lim\limits_{x\to a}\left([x-5]-[2x+2]\right)=0$, where $[\propto]$ denotes the greatest integer less than or equal to $\propto$ is equal to : **[JEE(Main)Jan.2023]**

a के सभी मानों, जिनके लिए $\lim\limits_{x\to a}\left([x-5]-[2x+2]\right)=0$ है, जहाँ $[\propto]$ महत्तम पूर्णांक $\le \propto$ है, का समुच्चय है : **[JEE(Main)Jan.2023]**

(A) (–7.5, –6.5) (B) (–7.5, –6.5] (C) [–7.5, –6.5] (D) [–7.5, –6.5)

Ans. A

Sol. $\lim\limits_{x\to a}\left([x-5]-[2x+2]\right)=0$

$\lim\limits_{x\to a}\left([x]-5-[2x]-2\right)=0$

$\lim\limits_{x\to a}\left([x]-[2x]\right)=7$

Let $a\in\left[n, n+\frac{1}{2}\right)$ then n-2 n=7

$n=-7$

Let $a\in\left[n+\frac{1}{2}, n+1\right)$ then

$n-(2n+1)=7$

$-n=8$

$n=-8$

$a \in \left[-7\frac{1}{2}, -7\right)$

but limit does not exists at a = –7.5

Hence $a \in (-7.5, -6.5)$

3. The value of $\lim\limits_{n\to\infty} \frac{1+2-3+4+5-6+.....+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ is : **[JEE(Main)Jan.2023]**

$\lim\limits_{n\to\infty} \frac{1+2-3+4+5-6+.....+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ का मान है : **[JEE(Main)Jan.2023]**

(A) $3\left(\sqrt{2}+1\right)$ (B) $\frac{3}{2}\left(\sqrt{2}+1\right)$ (C) $\frac{\sqrt{2}+1}{2}$ (D) $\frac{3}{2\sqrt{2}}$

Ans. B

Sol. $It_{n\to\infty} \frac{\sum\limits_{r=1}^{n}((3r-2)+(3r-1)-3r)}{\sqrt{2n^4+3n+1}-\sqrt{n^4+n+3}}$

$lt_{n\to\infty} \frac{\sum\limits_{r=1}^{n}3(r-1)}{\sqrt{2n^4+3n-1}-\sqrt{n^4+n+3}}$

$= lt_{n\to\infty} \frac{3\frac{n(n-1)}{2}\left(\sqrt{2n^4+3n-1}+\sqrt{n^4+n+3}\right)}{\left(2n^4+3n-1\right)-\left(n^4+n+3\right)}$

$= \frac{3}{2}(\sqrt{2}+1)$

4. Let x = 2 be a root of the equation $x^2 + px + q = 0$ and

$$f(x) = \begin{cases} \frac{1-\cos(x^2-4px+q^2+8q+16)}{(x-2p)^4} & , x \neq 2p \\ 0 & , x = 2p \end{cases}$$

Then $\lim\limits_{x\to 2p^+}[f(x)]$,

where $[\cdot]$ denotes greatest integer function, is : **[JEE(Main)Jan.2023]**

माना x = 2 समीकरण $x^2 + px + q = 0$ का एक मूल है और

$$f(x) = \begin{cases} \frac{1-\cos(x^2-4px+q^2+8q+16)}{(x-2p)^4} & , x \neq 2p \\ 0 & , x = 2p \end{cases}$$

है। तब $\lim\limits_{x\to 2p^+}[f(x)]$, जहाँ [·] महत्तम पूर्णांक फलन है का मान है : **[JEE(Main)Jan.2023]**

(A) -1 (B) 0 (C) 2 (D) 1

Ans. B

Sol. $$\lim_{x\to 2p^+}\left(\frac{1-\cos\left(x^2-4px+q^2+8q+16\right)}{\left(x^2-4px+q^2+8q+16\right)^2}\right)\left(\frac{\left(x^2-4px+q^2+8q+16\right)^2}{(x-2p)^2}\right)$$

$$\lim_{h\to 0}\frac{1}{2}\left(\frac{(2p+h)^2-4p(2p+h)+q^2+82+16}{h^2}\right)^2=\frac{1}{2}$$

Using L'Hospital's

$$\lim_{x\to 2p^+}[f(x)]=0$$

5. Let f, g and h be the real values functions defined on IR as

$$f(x)=\begin{cases}\dfrac{x}{|x|}, & x\neq 0\\ 1, & x=0\end{cases},\quad g(x)=\begin{cases}\dfrac{\sin(x+1)}{(x+1)}, & x\neq -1\\ 1, & x=-1\end{cases}$$

and $h(x)=2[x]-f(x)$, where $[x]$ is the greatest integer $\le x$.

Then the value fo $\lim\limits_{x\to 1}g\left(h(x-1)\right)$ is : **[JEE(Main)Jan.2023]**

माना f, g तथा h वास्तविक मान फलन है जो IR पर इस प्रकार परिभाषित है

$$f(x)=\begin{cases}\dfrac{x}{|x|}, & x\neq 0\\ 1, & x=0\end{cases},\quad g(x)=\begin{cases}\dfrac{\sin(x+1)}{(x+1)}, & x\neq -1\\ 1, & x=-1\end{cases}$$

एवं $h(x)=2[x]-f(x)$, जहाँ $[x]$ महत्तम पूर्णांक $\le x$ है। तब $\lim\limits_{x\to 1}g\left(h(x-1)\right)$ का मान है :

[JEE(Main)Jan.2023]

(A) $\sin(1)$ (B) 0 (C) -1 (D) 1

Ans. D

Sol. $\text{LHL}=\lim\limits_{k\to 0}\left(4(-k)\right),\ k>0$

$=\lim\limits_{k\to 0}g(-2+1)\qquad \because f(x)=-1\ \forall x<0$

$=g(-1)=1$

$\text{RHL}=\lim\limits_{k\to 0}g\left(h(k)\right),\ k>0$

$=\lim\limits_{k\to 0}g(-1),\qquad \because f(x)=1,\ \forall x>0$

= 1

6. $\lim\limits_{x\to\infty}\dfrac{\left(\sqrt{3x+1}+\sqrt{3x-1}\right)^6+\left(\sqrt{3x+1}-\sqrt{3x-1}\right)^6}{\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6}x^3$ **[JEE(Main)Jan.2023]**

(A) does not exist (B) is equal to 27

(C) is equal to 9 (D) is equal to $\dfrac{27}{2}$

$\lim\limits_{x\to\infty}\dfrac{\left(\sqrt{3x+1}+\sqrt{3x-1}\right)^6+\left(\sqrt{3x+1}-\sqrt{3x-1}\right)^6}{\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6}x^3$ **[JEE(Main)Jan.2023]**

(A) का अस्तित्व नहीं है (B) 27 के बराबर है

(C) 9 के बराबर है (D) $\dfrac{27}{2}$ के बराबर है

Ans. B

Sol. $\lim\limits_{x\to\infty}x^3\dfrac{\left(\sqrt{3x+1}+\sqrt{3x-1}\right)^6+\left(\sqrt{3x+1}-\sqrt{3x-1}\right)^6}{\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6}$

$(a+b)^6+(a-b)^6 = 2\,(T_0+T_2+T_4+T_6)$
$= 2\,({}^6C_0\,a^6 + {}^6C_2\,a^4b^2 + {}^6c_4\,a^2\,b^4 + {}^6C_6\,b^6)$

$$\lim_{x\to\infty}\frac{\left(2(3x+1)^3+30(3x+1)^2(3x-1)+30(3x+1)(3x-1)^2+2(3x-1)^3\right)x^3}{2x^6+30x^4\left(x^2-1\right)+30x^2\left(x^2-1\right)^2+\left(x^2-1\right)^3}$$

$$\lim_{x\to\infty}\frac{x^6\left[2\left(3+\frac{1}{x}\right)^3+30\left(3+\frac{1}{x}\right)^2\left(3-\frac{1}{x}\right)+30\left(3+\frac{1}{x}\right)\left(3-\frac{1}{x}\right)^2+2\left(3-\frac{1}{x}\right)^3\right]}{x^6\left[2+30\left(1-\frac{1}{x^2}+30\left(1-\frac{1}{x^2}\right)^2+2\left(1-\frac{1}{x^2}\right)^3\right]}$$

$$=\frac{2(3)^3+30(3)^3+30(3)^3+2(3)^3}{2+30+30+2}=(3)^3=27$$

7. Let $a_1, a_2, a_3, \ldots, a_n$ be n positive consecutive terms of an arithmetic progression. if $d > 0$ is its common difference, then $\lim\limits_{n\to\infty}\sqrt{\dfrac{d}{n}}\left(\dfrac{1}{\sqrt{a_1}+\sqrt{a_2}}+\dfrac{1}{\sqrt{a_2}+\sqrt{a_3}}+\ldots\ldots\ldots+\dfrac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$ is : **[JEE(Main)April,2023]**

माना $a_1, a_2, a_3, \ldots, a_n$ समांतर श्रेणी के धनात्मक क्रमागत n पद हैं। यदि सार्वन्तर $d > 0$ है, तब

$\lim\limits_{n\to\infty}\sqrt{\dfrac{d}{n}}\left(\dfrac{1}{\sqrt{a_1}+\sqrt{a_2}}+\dfrac{1}{\sqrt{a_2}+\sqrt{a_3}}+\ldots\ldots+\dfrac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$ का मान है : **[JEE(Main)April,2023]**

(A) 1 (B) 0 (C) $\sqrt{d}$ (D) $\dfrac{1}{\sqrt{d}}$

Ans. A

Sol. $\left(\dfrac{1}{\sqrt{a_1}+\sqrt{a_2}}+\dfrac{1}{\sqrt{a_2}+\sqrt{a_3}}+---+\dfrac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$

$=\left(\dfrac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1}+\dfrac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2}+\dfrac{\sqrt{a_4}-\sqrt{a_3}}{a_4-a_3}+---+\dfrac{\sqrt{a_n}-\sqrt{a_{n-1}}}{a_n-a_{n-1}}\right)$

$=\dfrac{1}{d}\left(\sqrt{a_n}-\sqrt{a_1}\right)$

$\therefore \lim\limits_{n\to\infty}\sqrt{\dfrac{d}{n}}\left(\dfrac{1}{\sqrt{a_1}+\sqrt{a_2}}+\dfrac{1}{\sqrt{a_2}+\sqrt{a_3}}+---+\dfrac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}\right)$

$=\lim\limits_{n\to\infty}\sqrt{\dfrac{d}{n}}\left(\dfrac{\sqrt{a_n}-\sqrt{a_1}}{d}\right)$

$=\lim\limits_{n\to\infty}\left(\dfrac{\sqrt{a_1+(n-1)d}-\sqrt{a_1}}{\sqrt{n}\sqrt{d}}\right)=\lim\limits_{n\to\infty}\left(\sqrt{\dfrac{a_1}{nd}+\left(1-\dfrac{1}{n}\right)}-\sqrt{\dfrac{a_1}{nd}}\right)=1$

8. $\lim\limits_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$ is equal to : **[JEE(Main)April,2023]**

$\lim\limits_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$ बराबर है : **[JEE(Main)April,2023]**

(A) 1 (B) $\sqrt{2}$ (C) 0 (D) $\dfrac{1}{\sqrt{2}}$

Ans. C

Sol. $P=\lim\limits_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right).....\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)$

Let

$2^{\frac{1}{2}}-2^{\frac{1}{3}}$ $\rightarrow$ Smallest

$2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}$ $\rightarrow$ Largest

Sandwich th.

$$\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n \le P \le \left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n$$

$$\begin{pmatrix}\text{lie b/w}\\ 0 \text{ and } 1\end{pmatrix}^n$$

$$\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n = 0$$

$$\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n = 0$$

$$\therefore P = 0$$

9. $\lim\limits_{x\to 0}\left(\left(\dfrac{\left(1-\cos^2(3x)\right)}{\cos^3(4x)}\right)\left(\dfrac{\sin^3(4x)}{\left(\log_e(2x+1)\right)^5}\right)\right)$ is equal to : **[JEE(Main)April,2023]**

$\lim\limits_{x\to 0}\left(\left(\dfrac{\left(1-\cos^2(3x)\right)}{\cos^3(4x)}\right)\left(\dfrac{\sin^3(4x)}{\left(\log_e(2x+1)\right)^5}\right)\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) 15 (B) 24 (C) 9 (D) 18

Ans. D

Sol. $$\lim_{x\to 0}\left[\frac{1-\cos^2 3x}{9x^2}\right]\frac{9x^2}{\cos^3 4x}\cdot\frac{\left(\frac{\sin 4x}{4x}\right)^3\times 64x^3}{\left[\frac{\ln(1+2x)}{2x}\right]^5\times 32x^5}$$

$$\lim_{x\to 0}\left(\frac{9}{1}\times\frac{1\times 64}{1\times 32}\right)=18$$

10. If $\alpha > \beta > 0$ are the roots of the equation $ax^2 + bx + 1 = 0$, and $\lim\limits_{x\to\frac{1}{\alpha}}\left(\dfrac{1-\cos\left(x^2+bx+a\right)}{2(1-\alpha x)^2}\right)^{\frac{1}{2}} = \dfrac{1}{k}\left(\dfrac{1}{\beta}-\dfrac{1}{\alpha}\right)$, then k is equal to : **[JEE(Main)April,2023]**

यदि समीकरण $ax^2 + bx + 1 = 0$ के मूल $\alpha > \beta > 0$ हैं तथा $\lim\limits_{x\to\frac{1}{\alpha}}\left(\dfrac{1-\cos\left(x^2+bx+a\right)}{2(1-\alpha x)^2}\right)^{\frac{1}{2}} = \dfrac{1}{k}\left(\dfrac{1}{\beta}-\dfrac{1}{\alpha}\right)$ है, तो k बराबर है:

[JEE(Main)April,2023]

(A) α (B) 2α (C) β (D) 2β

Ans. B

Sol. $\therefore ax^2+bx+1=a(x-\alpha)(x-\beta)\therefore \alpha\beta=\frac{1}{a}$

$\therefore x^2+bx+a=a(1-\alpha x)(1-\beta x)$

$$\therefore \lim_{x\to\frac{1}{\alpha}}\left\{\frac{1-\cos\left(x^2+bx+a\right)}{2(1-\alpha x)^2}\right\}^{\frac{1}{2}}=\lim_{x\to\frac{1}{\alpha}}\left\{\frac{1-\cos\left(a(1-\alpha x)(1-\beta x)\right)}{2\{a(1-\alpha x)(1-\beta x)\}^2}\cdot a^2(1-\beta x)^2\right\}^{\frac{1}{2}}$$

$$=\left[\frac{1}{2}\cdot\frac{1}{2}a^2\left(1-\frac{\beta}{\alpha}\right)^2\right]^{\frac{1}{2}}$$

$$=\frac{1}{2}\frac{1}{\alpha\beta}\left(1-\frac{\beta}{\alpha}\right)=\frac{1}{2}\left(\frac{1}{\alpha\beta}-\frac{1}{\alpha^2}\right)$$

$$=\frac{1}{2\alpha}\left(\frac{1}{\beta}-\frac{1}{\alpha}\right)=\frac{1}{k}\left(\frac{1}{\beta}-\frac{1}{\alpha}\right)$$

$\therefore k=2\alpha$

11. If $\lim\limits_{x\to0}\dfrac{e^{ax}-\cos(bx)-\dfrac{cxe^{-cx}}{2}}{1-\cos(2x)}=17$, then $5a^2+b^2$ is equal to : **[JEE(Main)April,2023]**

यदि $\lim\limits_{x\to0}\dfrac{e^{ax}-\cos(bx)-\dfrac{cxe^{-cx}}{2}}{1-\cos(2x)}=17$, है, तो $5a^2+b^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 72 (B) 76 (C) 64 (D) 68

Ans. D

Sol. $$\lim_{x\to0}\frac{e^{ax}-\cos\left(bx\right)-\dfrac{cxe^{-cx}}{2}}{\dfrac{\left(1-\cos2x\right)}{4x^2}\times4x^2}=17$$

On expansion,

$$\lim_{x\to0}\frac{\left(1+ax+\dfrac{a^2x^2}{2}\right)-\left(1-\dfrac{b^2x^2}{2}\right)-\dfrac{cx}{2}\left(1-cx\right)}{2x^2}=17$$

$$\lim_{x\to0}\frac{\left(a-\dfrac{c}{2}\right)x+x^2\left(\dfrac{a^2}{2}+\dfrac{b^2}{2}+\dfrac{c^2}{2}\right)}{2x^2}=17$$

For limit to be exist $a-\frac{c}{2}=0$

$a=\frac{c}{2}$

and $\frac{a^2+b^2+c^2}{4}=17$

$a^2 + b^2 + 4a^2 = 17 \times 4$

$5a^2 + b^2 = 68$

12. If $a=\lim\limits_{x\to0}\frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ and $b=\lim\limits_{x\to0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$, then the value of ab^3 is :

[JEE(Main)Jan.2024]

यदि If $a=\lim\limits_{x\to0}\frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ तथा $b=\lim\limits_{x\to0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$ हैं, तो ab^3 का मान है :

[JEE(Main)Jan.2024]

(A) 36 (B) 25 (C) 32 (D) 30

Ans. C

Sol. $a=\lim\limits_{x\to0}\frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$

$=\lim\limits_{x\to0}\frac{\sqrt{1+x^4}-1}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)}$

$=\lim\limits_{x\to0}\frac{x^4}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)\left(\sqrt{1+x^4}+\right.}$

Applying limit $a=\frac{1}{4\sqrt{2}}$

$b=\lim\limits_{x\to0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$

$=\lim\limits_{x\to0}\frac{(1-\cos x)(\sqrt{2}+\sqrt{1+\cos x})}{2-(1+\cos x)}$

$b=\lim\limits_{x\to0}(1+\cos x)(\sqrt{2}+\sqrt{1+\cos x})$

Applying limit $b=2(\sqrt{2}+\sqrt{2})=4\sqrt{2}$

Now, $ab^3 = \frac{1}{4\sqrt{2}} \times (4\sqrt{2})^3 = 32$

13. If $\lim\limits_{x\to 0} \frac{3+\alpha \sin x + \beta \cos x + \log_e(1-x)}{3\tan^2 x} = \frac{1}{3}$, then $2\alpha - \beta$ is equal to : **[JEE(Main)Jan.2024]**

यदि $\lim\limits_{x\to 0} \frac{3+\alpha \sin x + \beta \cos x + \log_e(1-x)}{3\tan^2 x} = \frac{1}{3}$ है, तो $2\alpha - \beta$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 7 (B) 1 (C) 5 (D) 2

Ans. C

Sol. $\lim\limits_{x\to 0} \frac{3+\alpha \sin x + \beta \cos x + \log_e(1-x)}{3\tan^2 x} = \frac{1}{3}$

$$\Rightarrow \lim_{x\to 0} \frac{3+\alpha\left[x - \frac{x^3}{3!} + \ldots\right] + \beta\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!}\ldots\right] + \left(-x - \frac{x^2}{2} - \frac{x^3}{3}\ldots\right)}{3\tan^2 x} = \frac{1}{3}$$

$$\Rightarrow \lim_{x\to 0} \frac{(3+\beta) + (\alpha - 1)x + \left(-\frac{1}{2} - \frac{\beta}{2}\right)x^2 + \ldots}{3x^2} \times \frac{x^2}{\tan^2 x} = \frac{1}{3}$$

$$\Rightarrow \beta + 3 = 0, \alpha - 1 = 0 \text{ and } \frac{-\frac{1}{2} - \frac{\beta}{2}}{3} = \frac{1}{3}$$

$$\Rightarrow \beta = -3, \alpha = 1$$

$$\Rightarrow 2\alpha - \beta = 2 + 3 = 5$$

14. $\lim\limits_{x\to 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$: **[JEE(Main)Jan.2024]**

(A) does not exist (B) is equal to 1

(C) is equal to 2 (D) is equal to –1

$\lim\limits_{x\to 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$: **[JEE(Main)Jan.2024]**

(A) का अस्तित्व नहीं है (B) 1 के बराबर है

(C) 2 के बराबर है (D) –1 के बराबर है

Ans. C

Sol. $\lim\limits_{x\to 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$

Since, it is an even function

$\Rightarrow$ LHL = RHL

$$RHL = \lim_{x\to 0^+} \frac{e^{2\sin x} - 2\sin x - 1}{x^2}$$

$$= \lim_{x\to 0^+} \frac{e^{2\sin x}(2\cos x) - 2\cos x}{2x}$$

$$= \lim_{x\to 0^+} = \left(\frac{e^{2\sin x} - 1}{2\sin x}\right)\left(\frac{2\sin x}{x}\right)\cos x$$

$= 1 \times 2 \times 1$

$= 2$

15. Let $f :\to R \to (0,\infty)$ be strictly increasing function such that $\lim_{x\to\infty} \frac{f(7x)}{f(x)} = 1$. Then, the value of of $\lim_{x\to\infty}\left[\frac{f(5x)}{f(x)} - 1\right]$ is equal to : **[JEE(Main)Jan.2024]**

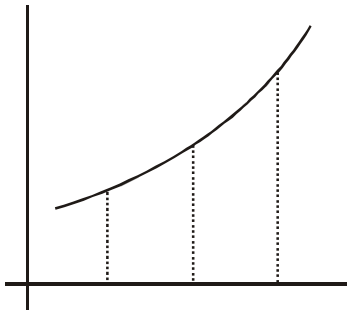
माना $f :\to R \to (0,\infty)$ निरंतर वर्धमान फलन है तथा $\lim_{x\to\infty} \frac{f(7x)}{f(x)} = 1$ है। तो $\lim_{x\to\infty}\left[\frac{f(5x)}{f(x)} - 1\right]$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{7}{5}$ (B) 0 (C) 1 (D) 4

Ans. B

Sol. $f(x) \le f(5x) \le f(7x), \forall x > 0$

$x < 5x < 7x$



$$1 \le \frac{f(5x)}{f(x)} \le \frac{f(7x)}{f(x)}$$

$$As\ x \to \infty, \frac{f(5x)}{f(x)} \to 1^+$$

$$\lim_{x\to\infty}\left[\frac{f(5x)}{f(x)} - 1\right] = 0$$

16. Let $f(x)=\begin{cases}x-1, x \text{ is even,}\\ 2x, \quad x \text{ is odd,}\end{cases}$ $x \in N$. If for some $a \in N, f\left(f\left(f(a)\right)\right)=21$, then $\lim\limits_{x \to a^-}\left\{\frac{|x|^3}{a}-\left[\frac{x}{a}\right]\right\}$, where [t] denotes the greatest integer less than or equal to t, is equal to : **[JEE(Main)Jan.2024]**

माना $f(x)=\begin{cases}x-1, x \text{ is even,}\\ 2x, \quad x \text{ is odd,}\end{cases}$ $x \in N$ यदि किसी $a \in N$ के लिए $f\left(f\left(f(a)\right)\right)=21$, तो $\lim\limits_{x \to a^-}\left\{\frac{|x|^3}{a}-\left[\frac{x}{a}\right]\right\}$, जहाँ [t] के सम्मान या कम महत्तम पूर्णांक दर्शाता है, बराबर होगा : **[JEE(Main)Jan.2024]**

(A) 225 (B) 144 (C) 121 (D) 169

Ans. B

Sol. $f(\underline{f(f(a))})=21$

$f(f(a))-1=21$

$f(f(a))=22$

$2f(a)=22$

$f(a)=11$

$a-1=11$

$\Rightarrow a=12$

$\lim\limits_{x \to 12}\left(\frac{|x|^3}{12}-\left[\frac{x}{12}\right]\right)$

$=144-0=144$

17. If $\lim\limits_{x \to 0}\frac{ax^2e^x-b\log_e(1+x)+cxe^{-x}}{x^2\sin x}=1$, then $16\,(a^2+b^2+c^2)$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि $\lim\limits_{x \to 0}\frac{ax^2e^x-b\log_e(1+x)+cxe^{-x}}{x^2\sin x}=1$ है, तो $16(a^2+b^2+c^2)$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 81

Sol. $\lim\limits_{x \to 0}\frac{ax^2e^x-b\log_e(1+x)+cxe^{-x}}{x^2\sin x}=1$

$$\lim_{x \to 0}\frac{ax^2\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}\ldots\right)-b\left(x-\frac{x^2}{2}+\frac{x^3}{3}+\ldots\right)+cx\left(1-x+\frac{x^2}{2!}\right)}{x^2\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}+\ldots\right)}$$

$$\lim_{x\to 0}\frac{(-b+c)x+x^2\left(a+\frac{b}{2}-c\right)+\left(a-\frac{b}{3}+\frac{c}{2}\right)x^3+\ldots}{x^3\left(1-\frac{x^2}{3!}+\frac{x^5}{5!}+\ldots\right)}$$

$-b+c=0$ $\qquad a+\frac{b}{2}-c=0$

$c=b$ $\qquad a+\frac{b}{2}-b=0$

$a=\frac{b}{2}$

$a-\frac{b}{3}+\frac{c}{2}=1$

$\frac{b}{2}-\frac{b}{3}+\frac{b}{2}=1$ $\qquad \Rightarrow \frac{3b-2b+3b}{6}=1$

$4b=6$ $\qquad c=\frac{3}{2}$ $\qquad a=\frac{b}{2}$

$b=\frac{3}{2}$ $\qquad a=\frac{3}{2\times 2}\Rightarrow\frac{3}{4}$

$a=\frac{3}{4}$

$16\left(a^2+b^2+c^2\right)$

$16\left(\frac{9}{16}+\frac{9}{4}+\frac{9}{4}\right)$

$16\left(\frac{9+36+36}{16}\right)\Rightarrow 81$

18. Let $\{x\}$ denote the fractional part of x and $f(x)=\frac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\{x\}-\{x\}^3}$, $x\neq 0$. If L and R respectively denotes the left hand limit of f(x) at $x=0$, then $\frac{32}{\pi^2}\left(L^2+R^2\right)$ is equal to _____.

[JEE(Main)Jan.2024]

माना $\{x\}$, x के भिन्नात्मक भाग को दर्शाता है तथा $f(x)=\frac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\{x\}-\{x\}^3}$, $x\neq 0$ है। यदि $x=0$ पर f(x) की

बाएँ पक्ष की सीमा तथा दाएँ पक्ष की सीमा क्रमशः L तथा R हैं, तो $\frac{32}{\pi^2}\left(L^2+R^2\right)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 18

Sol. $L=\lim\limits_{x\to0^-}\dfrac{\cos^{-1}\left(1-(x+1)^2\right)\sin^{-1}\left(1-(1+x)\right)}{(x+1)\left(1-(x+1)^2\right)},\{x\}=x+1$

$$=\lim_{x\to0^-}\frac{\cos^{-1}\left(-x^2-2x\right)\sin^{-1}(-x)}{(x+1)\left(-x^2-2x\right)}$$

$$=\frac{\cos^{-1}(0)}{1}\frac{1}{2}=\frac{\pi}{4}$$

$$R=\lim_{x\to0^+}\frac{\cos^{-1}\left(1-x^2\right)\sin^{-1}(1-x)}{x\left(1-x^2\right)},\{x\}=x$$

$$\lim_{x\to0^+}\frac{\dfrac{-1(-2x)\frac{\pi}{2}}{2\sqrt{1-\left(1-x^2\right)^2}}}{1}$$

$$\lim_{x\to0^+}\frac{\sqrt{2-x^2}}{1}\frac{\pi}{2}=\frac{\pi}{\sqrt{2}}$$

$$\frac{32}{\pi^2}\left(L^2+R^2\right)=\frac{32}{\pi^2}\left(\frac{\pi^2}{16}+\frac{\pi^2}{2}\right)=18$$

19. $\lim\limits_{n\to\infty}\dfrac{\left(1^2-1\right)(n-1)+\left(2^2-2\right)(n-2)+\cdots+\left((n-1)^2-(n-1)\right)\cdot1}{\left(1^3+2^3+\cdots+n^3\right)-\left(1^2+2^2+\cdots+n^2\right)}$ is equal to : **[JEE(Main)April,2024]**

$\lim\limits_{n\to\infty}\dfrac{\left(1^2-1\right)(n-1)+\left(2^2-2\right)(n-2)+\cdots+\left((n-1)^2-(n-1)\right)\cdot1}{\left(1^3+2^3+\cdots+n^3\right)-\left(1^2+2^2+\cdots+n^2\right)}$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

Ans. D

Sol. $\lim\limits_{n\to\infty}\dfrac{\left(1^2-1\right)(n-1)+\left(2^2-2\right)(n-2)+\ldots+\left(\left(n-1\right)^2-(n-1)\right)\times 1}{\left(1^3+2^3+\ldots..+n^3\right)\left(1^2+2^2+\ldots.+n^2\right)}$

$\text{Numerator}=\sum\limits_{r=1}^{n-1}\left(\left(r-1\right)^2-(r-1)\right)(n-r)$

$=\sum\limits_{r=1}^{n-1}(r-1)-(r-2)(n-r)$

$=\sum\limits_{r=1}^{n-1}-r^3+(n+3)r^2-(2+3n)r+2n$

We will take term with the greatest power of n

$=\dfrac{-1}{4}n^4+\dfrac{1}{3}+\dfrac{1}{3}n^4=\dfrac{1}{12}n^4$

$\text{Denominator}=\sum\limits_{r=1}^{n}r^3-\sum\limits_{r=1}^{n}r^2$

$=\left(\dfrac{n(n+1)^2}{2}\right)-\left(\dfrac{n(n+1)(2n+1)}{6}\right)$

Greatest power of n is $\dfrac{n^4}{4}$

$\lim\limits_{n\to\infty}\dfrac{\frac{1}{12}n^4}{\frac{n^4}{4}}=\dfrac{1}{3}$

20. Let a circle passing through (2, 0) have its centre at the point (h, k). Let (x_c, y_c) be the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$. If $h = \lim\limits_{c\to 1} x_c$ and $k = \lim\limits_{c\to 1} y_c$, then the equation of the circle is:

[JEE(Main)April,2024]

माना (2, 0) से होकर जाने वाले एक वृत्त का केन्द्र (h, k) है। माना रेखाओं $3x + 5y = 1$ तथा $(2 + c)x + 5c^2y = 1$ का प्रतिच्छेदन बिन्दु (x_c, y_c) है। यदि $h = \lim\limits_{c\to 1} x_c$ तथा $k = \lim\limits_{c\to 1} y_c$ है, तो वृत्त का समीकरण है : **[JEE(Main)April,2024]**

(A) $25x^2 + 25y^2 - 2x + 2y - 60 = 0$ (B) $5x^2 + 5y^2 - 4x + 2y - 12 = 0$

(C) $25x^2 + 25y^2 - 20x + 2y - 60 = 0$ (D) $5x^2 + 5y^2 - 4x - 2y - 12 = 0$

Ans. C

Sol. $3x + 5y = 1$

$(2 + c)x + 2c^2y = 1$

$(3c^2 - c - 2)x = (c^2 - 1)$

$$x = \frac{c^2-1}{3c^2-c-2} \text{ and } y = \frac{1}{5}\left(1 - \frac{3\left(c^2-1\right)}{3c^2-c-2}\right) = \frac{1-c}{5\left(3c^2-c-2\right)}$$

$$h = \lim_{c\to 1} \frac{(1-c)}{(1-c)}\left(\frac{1+c}{2+3c}\right) = \frac{2}{5}$$

$$k = \lim_{c\to 1} \frac{1-c}{5(c-1)(3c+2)} = \lim_{c\to 1} \frac{-1}{5(3c+12)} = -\frac{1}{25}$$

$$\text{Centre of circle} = \left(\frac{2}{5}, -\frac{1}{25}\right)$$

$$\text{Now radius of circle } \sqrt{\left(2-\frac{2}{5}\right)^2 + \left(\frac{1}{25}\right)^2} = \sqrt{\frac{64}{25} + \frac{1}{625}}$$

Equation of circle is

$$\left(x - \frac{2}{5}\right)^2 + \left(y + \frac{1}{25}\right)^2 = \frac{64}{25} + \frac{1}{625}$$

$$x^2 + y^2 - \frac{4}{5}x + \frac{2}{25}y + \frac{4}{25} + \frac{1}{625} = \frac{64}{25} + \frac{1}{625}$$

$25x^2 + 25y^2 - 20x + 2y + 4 = 64$

$25x^2 + 25y^2 - 20x + 2y - 60 = 0$

21. $\lim_{x\to 0} \frac{e - (1+2x)^{\frac{1}{2x}}}{x}$ is equal to : **[JEE(Main)April,2024]**

$\lim_{x\to 0} \frac{e - (1+2x)^{\frac{1}{2x}}}{x}$ बराबर है : **[JEE(Main)April,2024]**

(A) e (B) $\frac{-2}{e}$ (C) 0 (D) $e - e^2$

Ans. A

Sol. $\text{Lim}_{x\to 0} \frac{e - e^{\frac{1}{2x}\ell n(1+2x)}}{x}$

$$= \text{Lim}_{x\to 0}(-e)\frac{\left(e^{\frac{\ell n(1+2x)}{2x}-1}-1\right)}{x}$$

$$= \text{Lim}_{x\to 0}(-e)\frac{\ell n(1+2x)-2x}{2x^2}$$

$$= (-e)\times(-1)\frac{4}{2\times 2} = e$$

22. If $\lim\limits_{x\to 1}\frac{(5x+1)^{\frac{1}{3}}-(x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}}-(x+4)^{\frac{1}{2}}} = \frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$, where gcd(m, n) = 1, then 8m + 12n is equal to _____.

[JEE(Main)April,2024]

यदि $\lim\limits_{x\to 1}\frac{(5x+1)^{\frac{1}{3}}-(x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}}-(x+4)^{\frac{1}{2}}} = \frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$, जहाँ gcd(m, n) = 1 हैं, तो 8m + 12n बराबर है _____

[JEE(Main)April,2024]

Ans. 100

Sol. $\ell = \lim\limits_{x\to 1}\frac{(5x+1)^{\frac{1}{3}}-(x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}}-(x+4)^{\frac{1}{2}}}$

From : $\frac{0}{0}$, using L-H rule

$$\ell = \lim_{x\to 1}\frac{\frac{1}{3}\times 5(5x+1)^{-\frac{2}{3}}-\frac{1}{3}(x+5)^{-\frac{2}{3}}}{\frac{1}{2}\times 2(2x+3)^{-\frac{1}{2}}-\frac{1}{2}(x+4)^{-\frac{1}{2}}}$$

$$= \frac{\left(\frac{5}{3}-\frac{1}{3}\right)6^{-\frac{2}{3}}}{\frac{1}{2}5^{-\frac{1}{2}}} = \frac{8}{3}\times\frac{5^{\frac{1}{2}}}{6^{\frac{2}{3}}} = \frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$$

$\Rightarrow m = 8, n = 3$

$\Rightarrow 8m + 12n = 100$

23. Let a > 0 be a root of the equation $2x^2 + x - 2 = 0$. If $\lim\limits_{x\to\frac{1}{a}}\frac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{(1-ax)^2} = \alpha+\beta\sqrt{17}$ where

$\alpha, \beta \in Z$, then $\alpha + \beta$ is equal to _____. **[JEE(Main)April,2024]**

माना समीकरण $2x^2 + x - 2 = 0$ का एक मूल $a > 0$ है। यदि $\lim\limits_{x\to\frac{1}{a}} \frac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{\left(1-ax\right)^2} = \alpha+\beta\sqrt{17}$ है, जहाँ $\alpha, \beta \in Z$ है। तो $\alpha + \beta$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 170

Sol. $\therefore 2x^2 + x - 2 = 0$ has two roots where

$a = \frac{\sqrt{17}-1}{4}$ and another root is $\frac{-\sqrt{17}-1}{4}$

And $2 + x - 2x^2 = -2\left(x-\frac{1}{a}\right)\left(x+\frac{4}{\sqrt{17}+1}\right)$

Now $\lim\limits_{x\to\frac{1}{a}} \frac{16\left(1-\cos\left(2+x-2x^2\right)\right)}{\left(1-ax\right)^2}$

$\lim\limits_{x\to\frac{1}{a}} \frac{32\sin^2\left(2+x-2x^2\right)}{a^2\left(\frac{1}{a}-x\right)^2}$

$= \lim\limits_{x\to\frac{1}{a}} \frac{\left(x+\frac{4}{\sqrt{17}+1}\right)^2 32\cdot\left(\sin\left(\frac{1}{2}\cdot(-2)\right)\left(x-\frac{1}{a}\right)\left(x+\frac{4}{\sqrt{17}+1}\right)\right)^2}{a^2\left(\left(x-\frac{1}{a}\right)\left(x+\frac{4}{\sqrt{17}+1}\right)\right)^2}$

$= 2\cdot\left(\frac{1}{a}+\frac{4}{\sqrt{17}+1}\right)^2\cdot\left(\frac{4}{\sqrt{17}-1}\right)^2$

$= 2\left(\frac{4}{\sqrt{17}-1}+\frac{4}{\sqrt{17}+1}\right)^2\cdot\left(\frac{4}{\sqrt{17}-1}\right)^2$

$= \frac{17\times4}{18-2\sqrt{17}} = \frac{68}{9-\sqrt{17}}$

$= 17\left(9+\sqrt{17}\right)$

$\alpha + \beta = 170$

24. If $\alpha = \lim\limits_{x\to 0^+}\left(\dfrac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}}\right)$ and $\beta = \lim\limits_{x\to 0}(1+\sin)^{\frac{1}{2}\cot x}$ are the roots of the quadratic equation $ax^2 + bx - \sqrt{e} = 0$, then $12\log_e(a+b)$ is equal to _____. **[JEE(Main)April,2024]**

यदि $\alpha = \lim\limits_{x\to 0^+}\left(\dfrac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}}\right)$ तथा $\beta = \lim\limits_{x\to 0}(1+\sin)^{\frac{1}{2}\cot x}$ द्विघात समीकरण $ax^2 + bx - \sqrt{e} = 0$ के मूल हैं, तो $12\log_e(a+b)$ बराबर है _____. **[JEE(Main)April,2024]**

Ans. 6

Sol. $\alpha = \lim\limits_{x\to 0^+}\dfrac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\left(\sqrt{\tan x} - \sqrt{x}\right)}$

$= \lim\limits_{x\to 0}\dfrac{e^{\sqrt{x}}\left(e^{\sqrt{\tan x}-\sqrt{x}} - 1\right)}{\left(\sqrt{\tan x} - \sqrt{x}\right)} = 1$

$\beta = \lim\limits_{x\to 0}(1+\sin x)^{\frac{1}{2}\cot x} = \lim\limits_{x\to 0} e^{(\sin x)\left(\frac{1}{2}\cot x\right)}$

$= \lim\limits_{x\to 0} e^{\frac{1}{2}\cos x} = e^{\frac{1}{2}}$

Product of roots $= \sqrt{e} = \dfrac{-\sqrt{e}}{a} \Rightarrow a = -1$

Sum of roots $= \dfrac{-b}{a} = 1 + \sqrt{e}$

$= b = \sqrt{e} + 1$

$\Rightarrow 12\ell n(a+b) = 12\ell n\left(\sqrt{e} + 1 - 1\right) = 12\ell n\left(e^{\frac{1}{2}}\right) = 6$